Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/14/2012
PI Name:	Swenson, Douglas Ph.D.		
Project Title:	Systematic Investigation of Organized Elongated Pore Formation in Invariant Liquid to Solid Metal Plus Gas Transformations		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	49931-1200	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	09/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 9/30/2014, per NSSC information (Ed., 9/20/13)		
Key Personnel Changes/Previous PI:	No Changes of Key Personnel		
COI Name (Institution):	Sanders, Paul (Michigan Technological University)		
Grant/Contract No.:	NNX10AV37G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Porous metals are metals specially processed to contain a large volume fraction of pores (voids). They possess many properties similar to solid metals, such as high strength and stiffness, but have much lower densities. They are well-suited to applications such as aerospace vehicles, vehicle crush zones, and sound absorption panels. Many of their properties are sensitive to pore geometry (fraction, size, distribution, etc.). Recently, one production process has been found that provides great promise for controlling pore geometry. It involves solidifying liquid metals containing dissolved gases that are released upon solidification, resulting in pores that are parallel tubes of uniform diameter and spacing. This is an example of cooperative growth, where two different phases form from a parent phase in a geometry that allows the growth of one phase to assist the simultaneous growth of the other. Cooperative growth occurs only under limited ranges of experimental conditions. Liquid = Solid + Gas cooperative growth has had very little scientific		

Task Description:	investigation, so the relationships between processing conditions and cooperative growth geometry remain unclear. The goal of this project is to conduct a systematic study of the Liquid = Solid + Gas transformation in metal-gas systems to determine how processing parameters, such as solidification rate and gas pressure, affect pore geometry. This work will identify process windows for producing uniform porous metals, and provide data that can be used to tailor cooperative growth models to this type of transformation. There is also limited evidence that pore geometry is affected by gravitational forces. In the future, the research described here could be replicated in space, and the effects of gravity on Solid + Gas cooperative growth could be determined. All these efforts would lead to a better understanding of coupled Solid + Gas growth, and may point to new strategies for fabricating improved porous metals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are numerous potential terrestrial benefits that may result from conducting the research outlined for this grant. In practical terms, it has been shown that porous metals and alloys with aligned, tubular pores have mechanical properties superior to porous metals in which pore size is random and/or pore shape is nominally spherical. In fact, it has been shown that in some instances metals with aligned pores are as strong or even stronger in the direction of alignment than the same metal in the absence of porosity. High strength, porous metals would be useful for a wide variety of applications, including lightweight and therefore more fuel efficient vehicles and superior crush-proof panels to enhance safety in vehicles and military equipment. From a scientific standpoint, the liquid = metal + gas reaction is similar to many other important solidification processes that do not involve gases, but has not been studied systematically. By understanding this process, it may prove possible to enhance current understanding of the other, more well studied phenomena.
Task Progress:	Four major tasks have been performed during Year Two of this project. i.) Continued experimentation on forming aligned porous metal structures, utilizing our modified vacuum casting system. ii.) Thermodynamic modeling of the Liquid Goes to Metal + Gas process. iii.) Continued construction of a new, high pressure directional solidification furnace that is dedicated to this project for its remainder. iv.) Construction of a low temperature directional solidification furnace to be used with gas-saturated, transparent organic materials. Below, the progress made in each of these individual tasks is discussed. PROGRESS AND EXPERIMENTAL RESULTS USING THE VACUUM CASTING SYSTEM Our vacuum casting furnace was reconfigured to allow the simultaneous introduction of a soluble gas and an insoluble gas (in our case, hydrogen and argon, respectively) in amounts and at pressures controlled by mass flow controllers and a dynamic evacuation system. It is a clean system capable of evacuation to approximately 10-5 Torr prior to melting and gas introduction, and can render metals molten with melting temperatures up to about 1550 degrees C. The pressure and temperature ranges of this system are very similar to those that could be utilized during potential future microgravity studies aboard the International Space Station (ISS), using the Solidification and Quenching Furnace (SQF) or Low Gradient Furnace (LGF). We have focused our studies on producing porous copper under varying pressure conditions, and to date have successfully cast dozens of specimens with porosities as high as 50%. We are currently analyzing our data in order to correlate processing conditions with the resulting porous microstructures, and are obtaining additional useful information, such as the cooling rate of the material as a function of its position in the casting mold. PROGRESS IN THERMODYNAMIC MODELING OF THE LIQUID GOES TO SOLID + GAS TRANSFORMATION We have developed a formal thermodynamic description of the Liquid Goes to Solid + Gas phase transformation, utilizing as a starting point the classical thermodynamic treatment of “dilute solutions”. To date, such analyses reported in the literature include some fundamental misconceptions pertaining to the effect of an “inert” or nonreactive gas pressure on chemical equilibrium. “Dilute solution” refers to the situation where the solute element (here, hydrogen) has a very low solubility (<< 1 at.%) in the solvent element (in this case, the metal). This work is similar to extensive analyses first developed in the 1950s through 1960s for the application of steelmaking, since small quantities of various elements have a great impact upon the resulting quality of steel if these impurities are not removed. We have focused on one special case of involving the dissociation and dissolution of hydrogen molecules into discrete hydrogen atoms once the hydrogen goes into solution in a liquid or solid. Hence, one mole of H2 gas, upon dissolution, becomes two moles of H atoms in solution. This special case is referred to as Sieverts’ Law, after the German chemist who first observed the phenomenon. Using this type of treatment, we are able to calculate Metal-Hydrogen binary phase systems as functions of hydrogen pressure and temperature. Important results of this analysis are: i.) as the hydrogen pressure increases, the solubility of hydrogen atoms increases in both the solid and liquid phases; ii.) as the hydrogen pressure increases, the temperature of the Solid + Liquid + Gas equilibrium decreases, and iii.) as the hydrogen pressure increases, the compositions of solid and liquid that coexist with hydrogen gas also increase. We are also able to show that the presence of “extra” pressure from a second gas like argon has negligible effect upon a Metal-Hydrogen phase diagram. It does, however, affect the total amount of porosity that develops in a cast structure by “compressing” the pores as they develop. PROGRESS IN CONSTRUCTION OF A HIGH TEMPERATURE PRESSURE VESSEL FURNACE Our current casting system has two main experimental drawbacks. First, gas pressures may only be varied over a limited range. Second, the solidification rate of the liquid alloy cannot be controlled easily. We believe both of these variables are important in terms of resulting microstructure, and that if we could extend their variations we could more easily study and understand the Liquid -->Solid + Gas transformation. We were able to achieve some additional funding, outside of this grant, to build another furnace with enhanced capabilities. The new vessel will allow total gas pressures within the system to range from 10-9 to nearly 3 MPa. Soluble and inert gases may be introduced independently from each other. Ingots of up to about 30 cm in height may be solidified directionally by lowering them out of the induction furnace using a magnetic transport system, with additional active water cooling. Velocity control with the magnetic

	transport shaft ranges from a few microns to a few millimeters per second, allowing orders of magnitude control of solidification rates. The high pressure directional solidification furnace is nearing completion. It is anticipated that the furnace will be functional near the end of the Fall 2012 semester. PROGRESS IN CONSTRUCTION OF A TRANSPARENT ORGANICS DIRECTIONAL SOLIDIFICATION FURNACE Last Summer, we also obtained additional, outside funds to build an auxiliary, low temperature solidification furnace that should greatly augment the work done under this project. This furnace has been designed to study directional solidification of low melting temperature, transparent organic materials that have been saturated with gases while in the molten state. This new furnace will allow us to observe in real time the Liquid Goes to Solid + Gas process, i.e., we can literally watch the process happen through clear quartz tubing and can also see through the liquid, solid and gas phases. This furnace is also near completion. It is anticipated that this furnace will be finished during the latter part of the 2012 Fall Semester, and testing will begin thereafter, utilizing succinonitrile and hydrogen or nitrogen as an initial transparent organic material test system for studying the Liquid Goes to Solid + Gas transformation.
Bibliography Type:	Description: (Last Updated: 11/14/2012)
Abstracts for Journals and Proceedings	Swenson D, Sanders P, Lifer A, Ranck H. "Thermodynamics of Metal-Gas Eutectic Systems and its Application to Aligned Porous Metal Fabrication." 50th AIAA Aerospace Sciences Meeting, including the New Horizons Forum and Aerospace Exposition, Nashville, Tennessee, January 9-12, 2012. 50th AIAA Aerospace Sciences Meeting, including the New Horizons Forum and Aerospace Exposition, Nashville, Tennessee, January 9-12, 2012. Oral presentation by D. Swenson. , Jan-2012
Abstracts for Journals and Proceedings	Swenson D, Sanders P, Lifer A, Ranck H. "Thermodynamics of Metal-Gas Eutectic Systems and its Application to Aligned Porous Metal Fabrication." TMS 2012. 141st Annual Meeting, The Minerals, Metals and Materials Society, Orlando, FL, March 11-15, 2012. Presentation by D. Swenson. TMS 2012. 141st Annual Meeting, The Minerals, Metals and Materials Society, Orlando, FL, March 11-15, 2012. Conference Program, p. 183. http://www.tms.org/Meetings/Annual-12/PDFs/TMS2012FinalProgramWEBwithAbstracts.pdf , Mar-2012